

Maximilien Le Cleï

PhD student @ Mila (world's largest modern AI academic research center)

Previously: Lead Applied Research Scientist @ D-BOX Technologies (haptic motion technology)

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Professional Experience

PhD Student Researcher @ Mila - Quebec AI Institute, Toronto (Remote), Canada Oct 2024 - Present

- Human agentic & brain behaviour imitation (focus: continual learning & out-of-distribution generalization), GECCO 2024 publication

Lead Applied Research Scientist @ D-BOX Technologies, Toronto (Remote), Canada Jan 2024 - Jun 2025

- Convinced leadership and successfully led the company in applying generative AI to imitate D-BOX haptic artist tracks (generative modeling of 1D signals – chair actuators displacement – with audio+video conditional diffusion transformers | state space models)
- Strategized with leadership to create new solution offerings that are predicted to expand upon D-BOX's ~40M dollar revenue generating customer base, with generous newfound profit margins given AI-centricity
- Trained billion parameter models on terabytes of data. Ran carefully-crafted, systematic scaling experiments over task complexity, training regimes, architectural choices, input modalities, model size & dataset size

Research Scientist @ Montreal Geriatric Institute, Toronto (Remote), Canada Sep 2021 - Aug 2023

- Led the development of a strongly automated and highly collaborative machine learning (deep learning + evolutionary machine learning) training workspace to speed up and strengthen the experiments of present machine learning researchers
- Technical contribution: audio model alignment with fMRI data to improve generalization, published in Imaging Neuroscience 2025
- Technical advising: large language model (LLM) fine-tuning + visual language model (VLM) for fMRI brain data alignment
- Evolutionary machine learning for human video game behaviour cloning & modern reinforcement learning benchmarks
- First-author publications at major machine learning conferences : ICLR ALOE workshop 2022, GECCO 2022 & 2023

MSc Student Researcher @ Mila - Quebec AI Institute, Montréal, Canada Jan 2019 - Aug 2021

- Research and development of novel evolutionary machine learning algorithms for imitation and reinforcement learning
- Developed i) a highly scalable and distributed open-source general purpose evolutionary machine learning library ii) a novel high-level imitation learning paradigm and iii) extremely compute efficient artificial neural networks

Applied Research Scientist Intern @ Capbeast, Montréal, Canada Jan 2018 - Apr 2018

- Early-stage application of modern deep reinforcement learning techniques to industrial embroidery problems
- Developed embroidery virtual environments of various complexity/fidelity & applied state-of-the-art deep reinforcement learning algorithms (DQN, Prioritized experience replay & Dueling Double-Q-Learning) to prototype highly capable embroidering agents

Education

University of Montréal, Montréal, Canada

Doctor of Philosophy - PhD, Computer Science (Artificial Intelligence) Sep 2023 - Present

Master of Science - MSc, Computer science (Artificial Intelligence) Sep 2018 - Aug 2021

Concordia University, Montréal, Canada

Bachelor of Computer Science - BCompSc, Computer Science (Computer Systems) Sep 2015 - Apr 2018

Skills

Topics: generative AI, deep learning, evolutionary machine learning, high performance computing, imitation learning, reinforcement learning, continual learning, out-of-distribution generalization, computer vision, NLP, robotics, MLOps, CI/CD

Technical: Python, PyTorch, Lightning, JAX, Tensorflow, MPI, HuggingFace, Weights & Biases, Bash, Slurm, Hydra, Submitit, Docker, GCP, GitHub Actions, Git, Pre-commit, Sphinx, Beartype, Devcontainers, Pytest, Jaxtyping, Claude Code

Open-source contributions: google/evojax (scalable evolutionary machine learning), ROCm/TheRock (AMD ROCm/PyTorch)